

# July 2025, CSE 106

## Online on Arrays

Section: C1+C2

Time: 40 minutes

Given a sorted array of integers  $A$ , and a target integer  $t$ , your goal is to find any pair of **unique indices**  $(i, j)$  such that  $A[i] + A[j] == t$ . You must implement this in  $O(n)$  time without using any additional arrays. If there are multiple such pairs, you may find any one of them.

You are free to tackle the problem however you like, but that approach will earn at most 60%. To receive full marks, you need to meet the stated time and space constraints.

=== Test 1 ===

$A = [1, 2, 4, 7, 11, 15]$

$T = 15$

Result:

$(2, 4)$

Explanation:  $A[2] + A[4] = 4 + 11 = 15$

=== Test 2 ===

$A = [-10, -3, -1, 0, 2, 4, 9]$

$t = -1$

Result:

$(0, 6)$

=== Test 3 ===

$A = [2, 2, 2, 2, 2]$

$t = 4$

Result:

$(0, 4)$

=== Test 4 ===

$A = [-5, -2, 0, 3, 9, 14]$

$t = 100$

Result:

No pairs found.

=== Test 5 ===

$A = [-2147483648, -1, 0, 1, 2147483647]$

$t = -2147483649$

Result:

$(0, 1)$